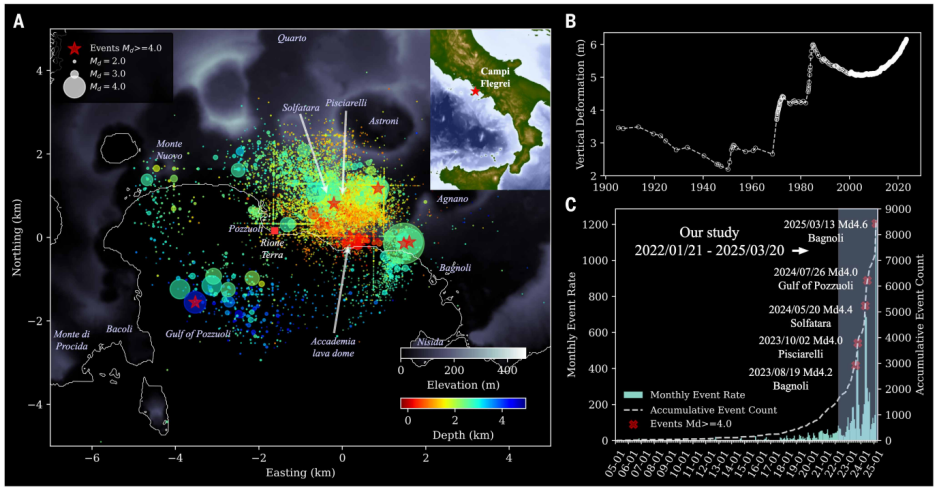
Figure 1: Seismicity and ground displacement

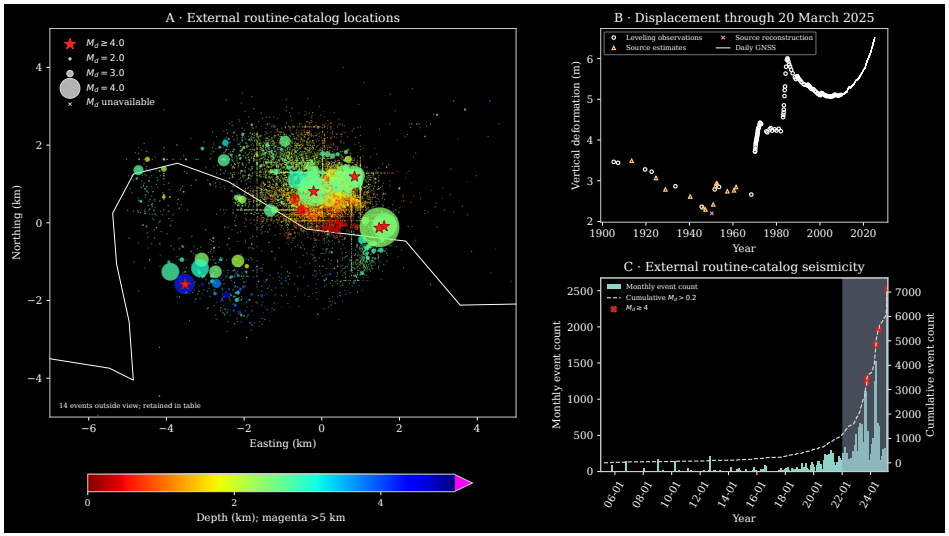
Published reference



Tan et al. (2025), tanmodel.pdf, p. 2.
DOI: 10.1126/science.adw9038.

Independent result

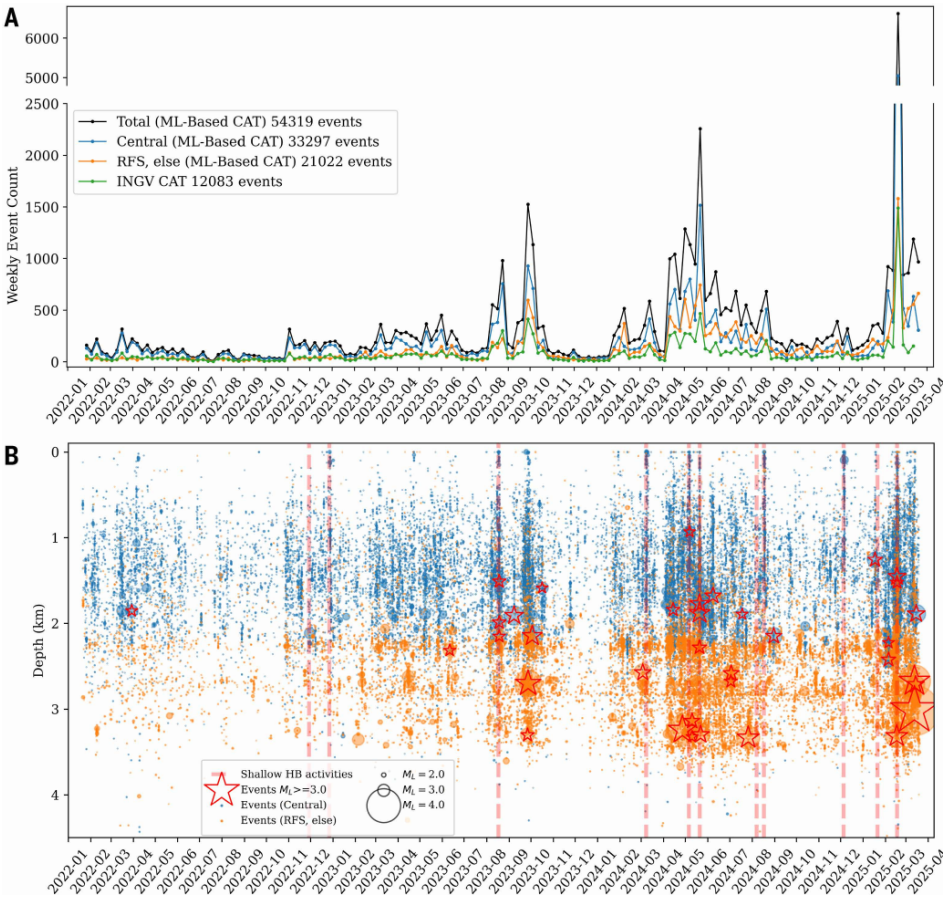
Panel 1A–C — partially reproduced



External routine-catalog context and source-labeled leveling/GNSS observations. These earthquakes are not independent detections. Historical snapshot and datum differences remain.

Figure 2: Temporal evolution of seismicity

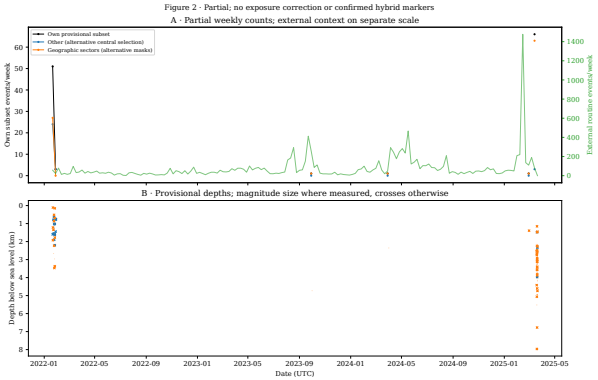
Published reference



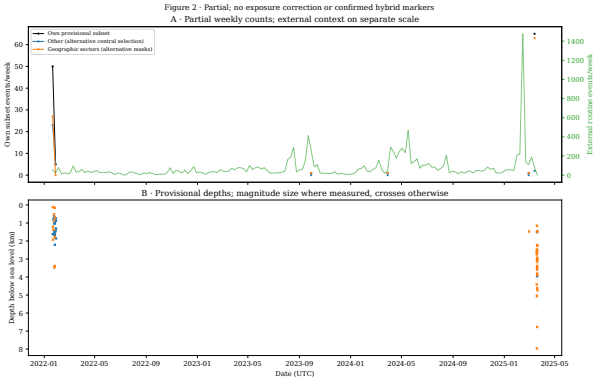
Tan et al. (2025), tanmodel.pdf, p. 3.
DOI: 10.1126/science.adw9038.

Independent result

Panel 2; provisional numerical counterpart — partially reproduced



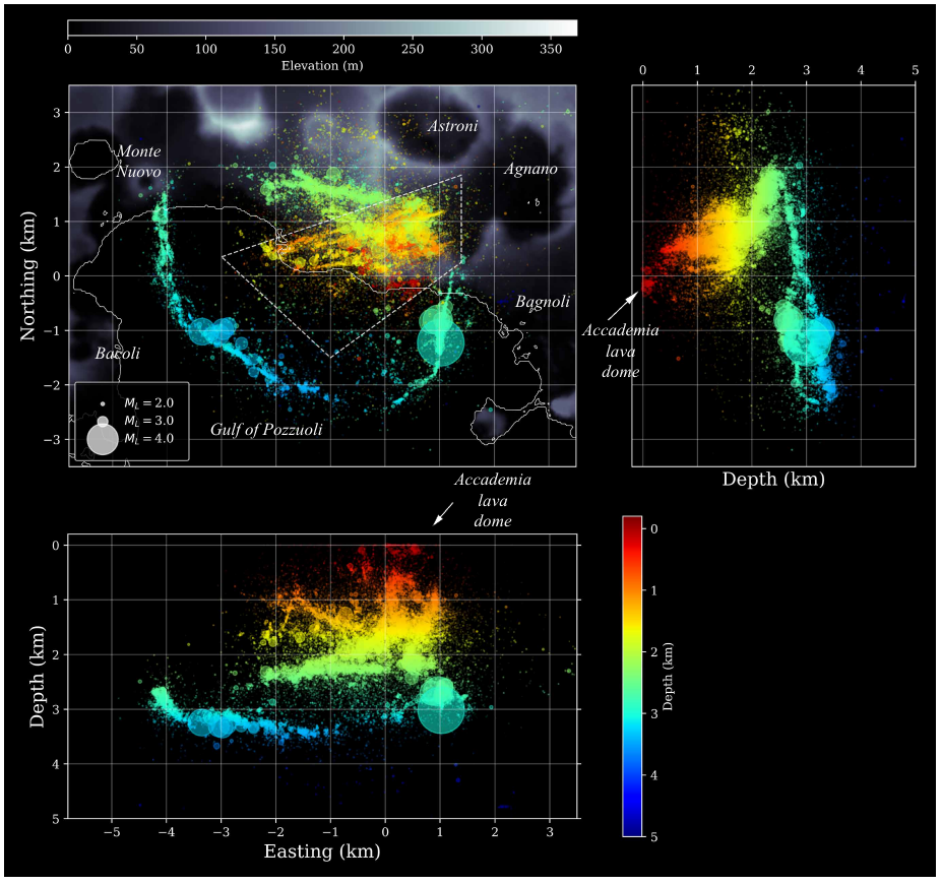
Panel Latest extended cohort: independently rerun, provisional — partially reproduced



Provisional weekly counts and depths from two overlapping own subsets. Sparse exposure is not a full-period rate measurement; spatial masks differ. Routine context has a separate axis.

Figure 3: Spatial distribution of seismicity

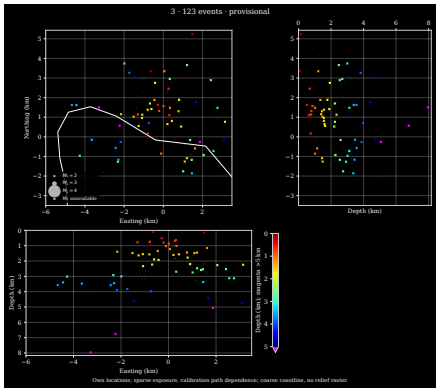
Published reference



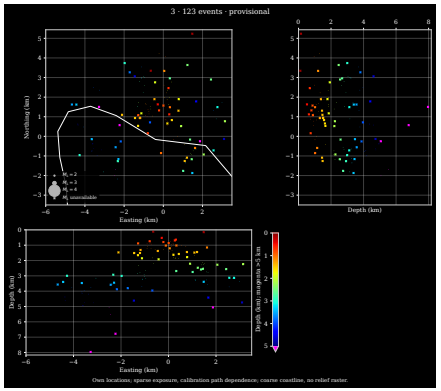
Tan et al. (2025), tanmodel.pdf, p. 4.
DOI: 10.1126/science.adw9038.

Independent result

Panel 3; provisional numerical counterpart — partially reproduced



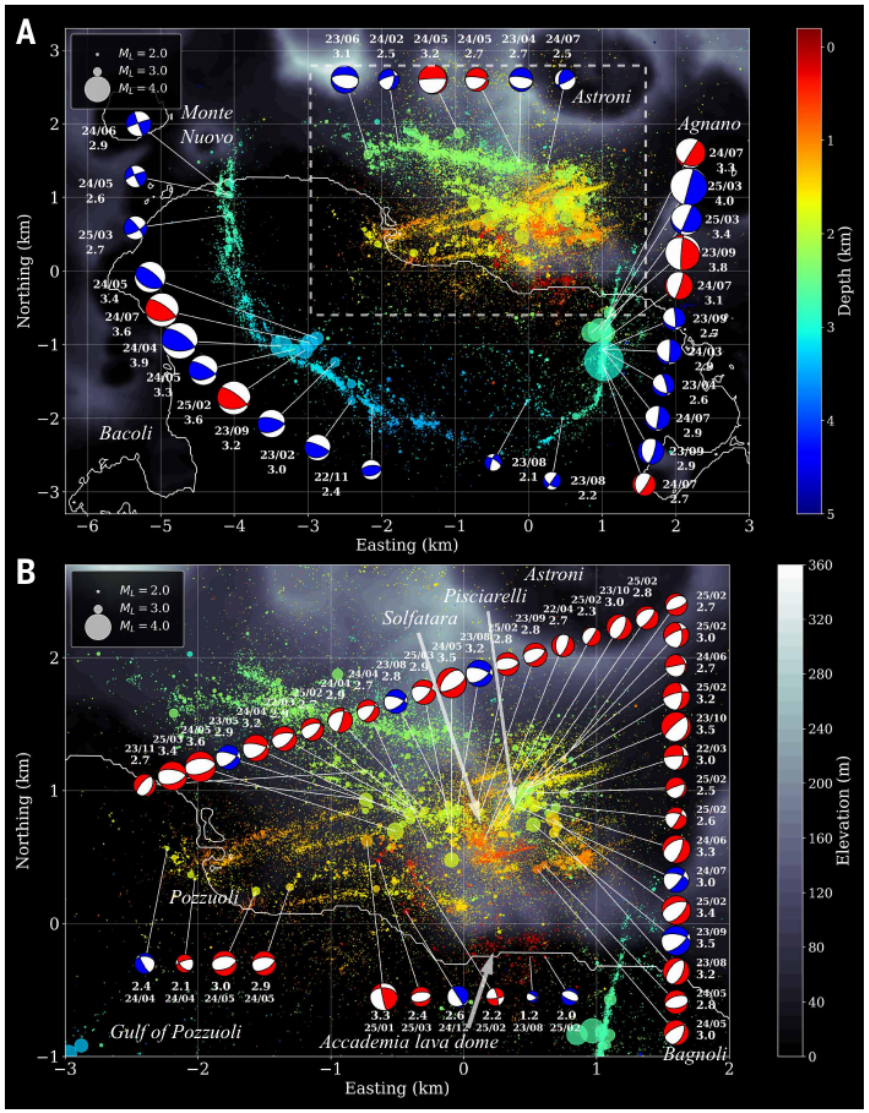
Panel Latest extended cohort: independently rerun, provisional — partially reproduced



Two separately processed 123-event trial subsets. Calibration sensitivity prevents accepting fault geometry. Own magnitudes set sizes where measured; crosses mark missing magnitudes. Depth colors retain over-range values.

Figure 4: Focal mechanisms of larger events

Published reference



Tan et al. (2025), tanmodel.pdf, p. 5.
DOI: 10.1126/science.adw9038.

Independent result

Missing: 4A, 4B — not reproduced

SKHASH was executed: 76 automated first-motion measurements on 8 own events; 35 passed the provisional sign/SNR screen. Seven events lack eight stable polarities; the remaining event has a 186.158-degree gap, exceeding the unchanged 90-degree gate. No mechanism is returned. Reviewed polarities, S/P constraints and independent gCAP/FK processing remain missing; no beachballs can be plotted.

Figure S1: Recorded station availability

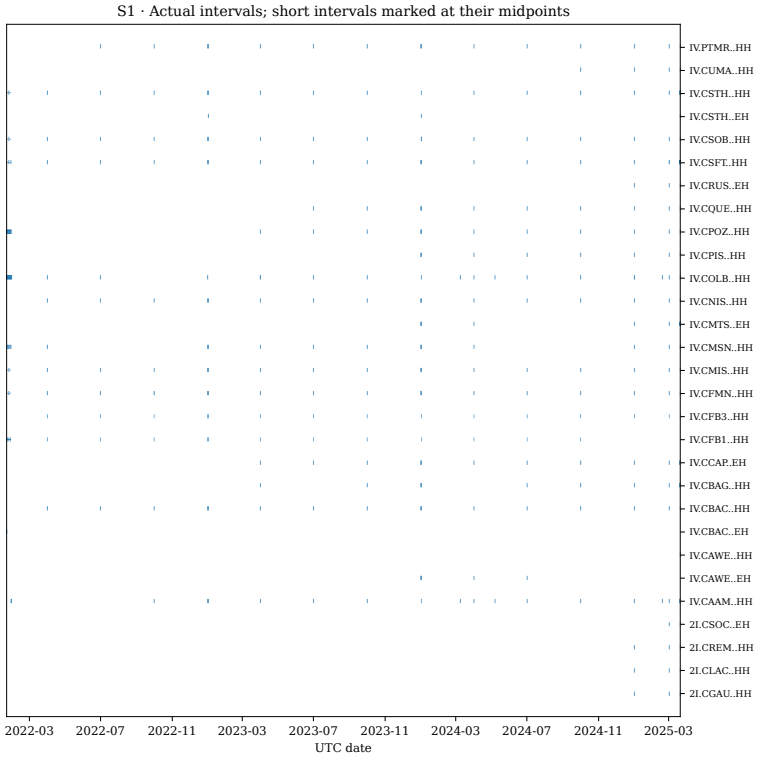
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 12.
DOI: 10.1126/science.adw9038.

Independent result

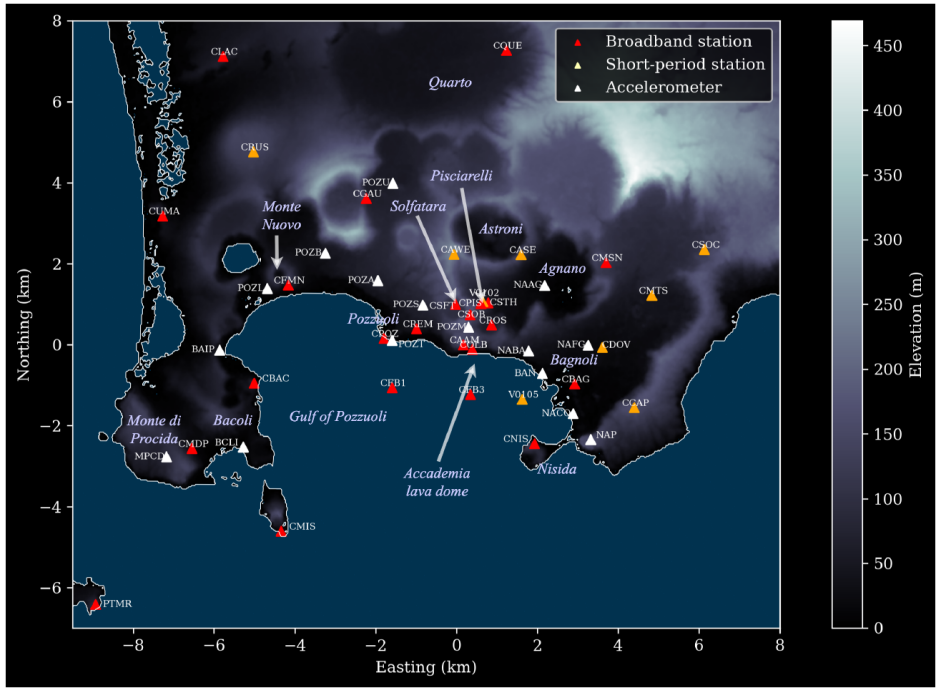
Panel S1: latest actual processed intervals — partially reproduced



997 files, 26 station names and 8706.970 union channel-hours. Short interval markers do not imply full-period coverage. Earlier calibration/downstream cohort remains separately numbered.

Figure S2: Station locations

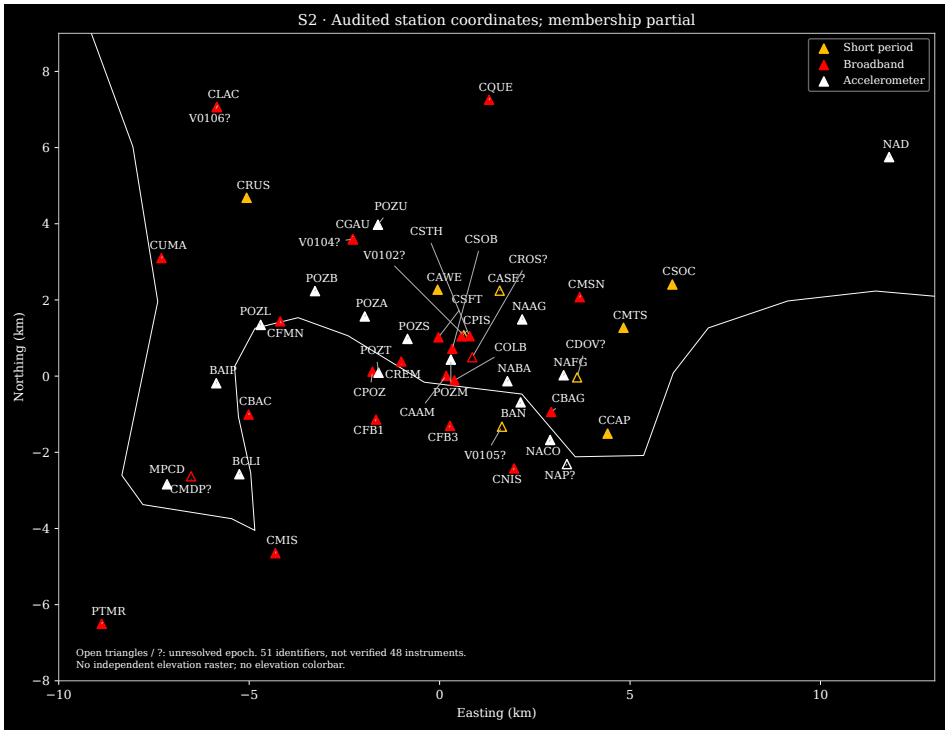
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 13.
DOI: 10.1126/science.adw9038.

Independent result

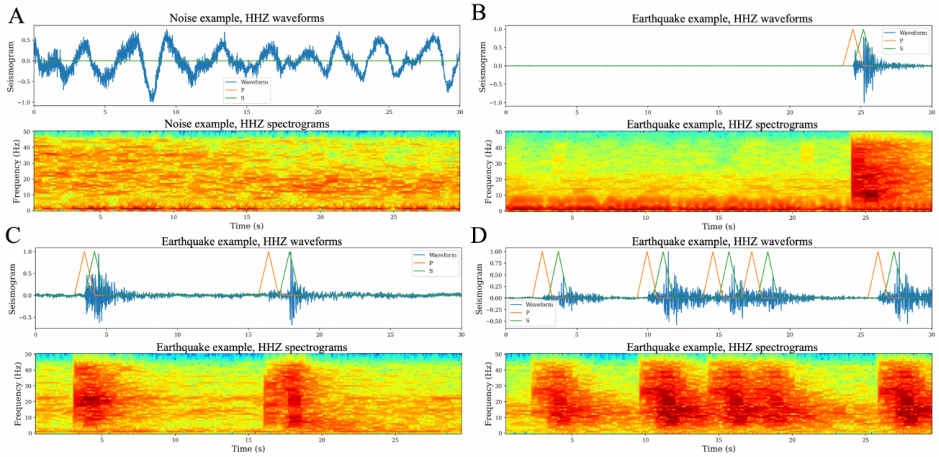
Panel S2 — partially reproduced



Partial station inventory: 51 released identifiers audited against official study-period response epochs. Filled triangles: 42 official coordinate matches; open triangles: 9 released coordinates with unresolved epochs or channel codes. Two coincident-coordinate pairs remain distinct identifiers. No verified geographic outlier. Local WGS84 projection and clipped coastline; the original 48-instrument membership is not established. Colors distinguish broadband (red), short period (yellow) and accelerometer (white) instruments; no independent relief raster is available.

Figure S3: Training examples

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 14.

DOI: 10.1126/science.adw9038.

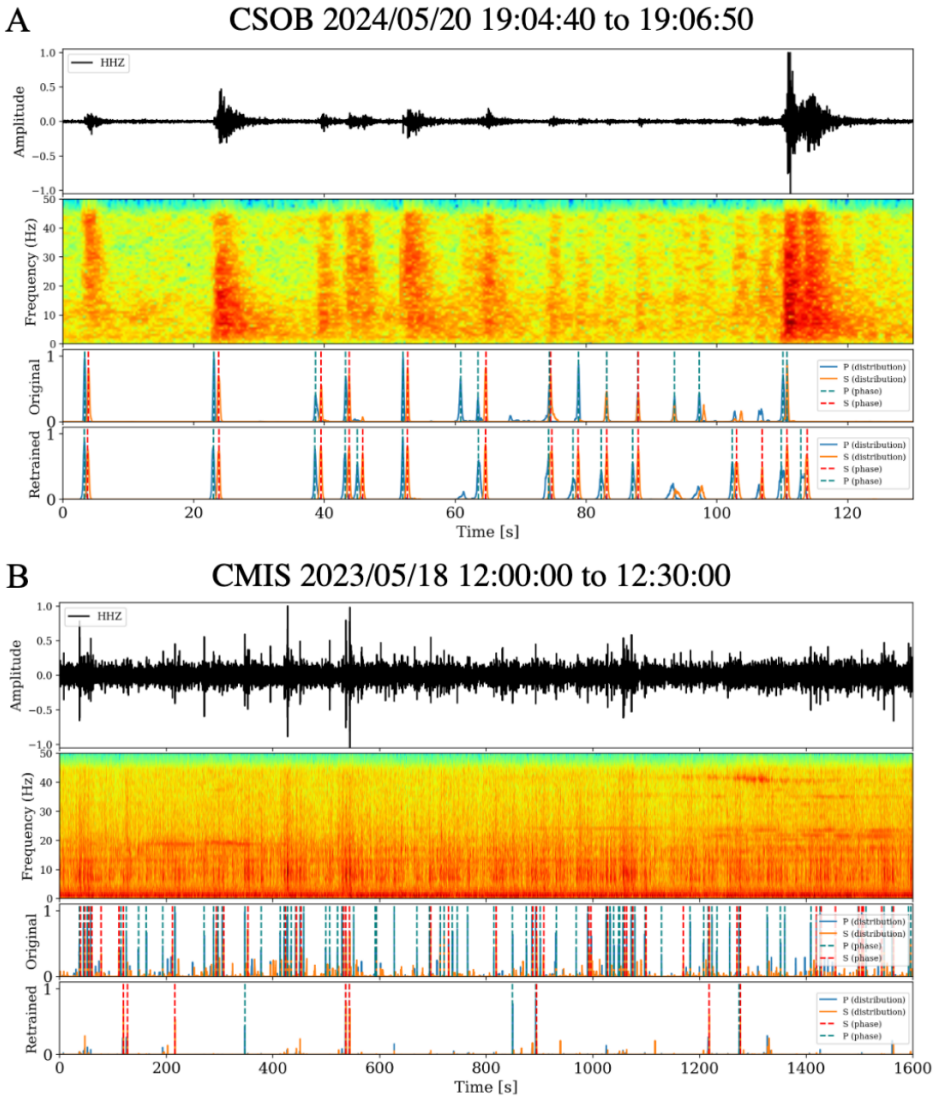
Independent result

Missing: S3A, S3B, S3C, S3D — not reproduced

Panels A–D require the specified training traces, annotations and augmentation examples. These were not reconstructed; downstream use of trained weights is not retraining.

Figure S4: PhaseNet benchmark examples

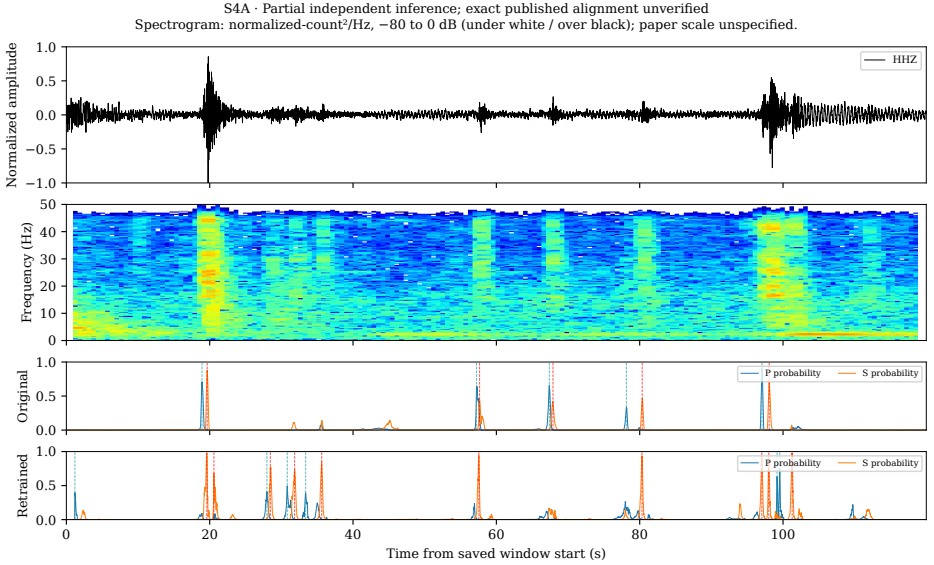
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 15.
DOI: 10.1126/science.adw9038.

Independent result

Panel S4A — partially reproduced



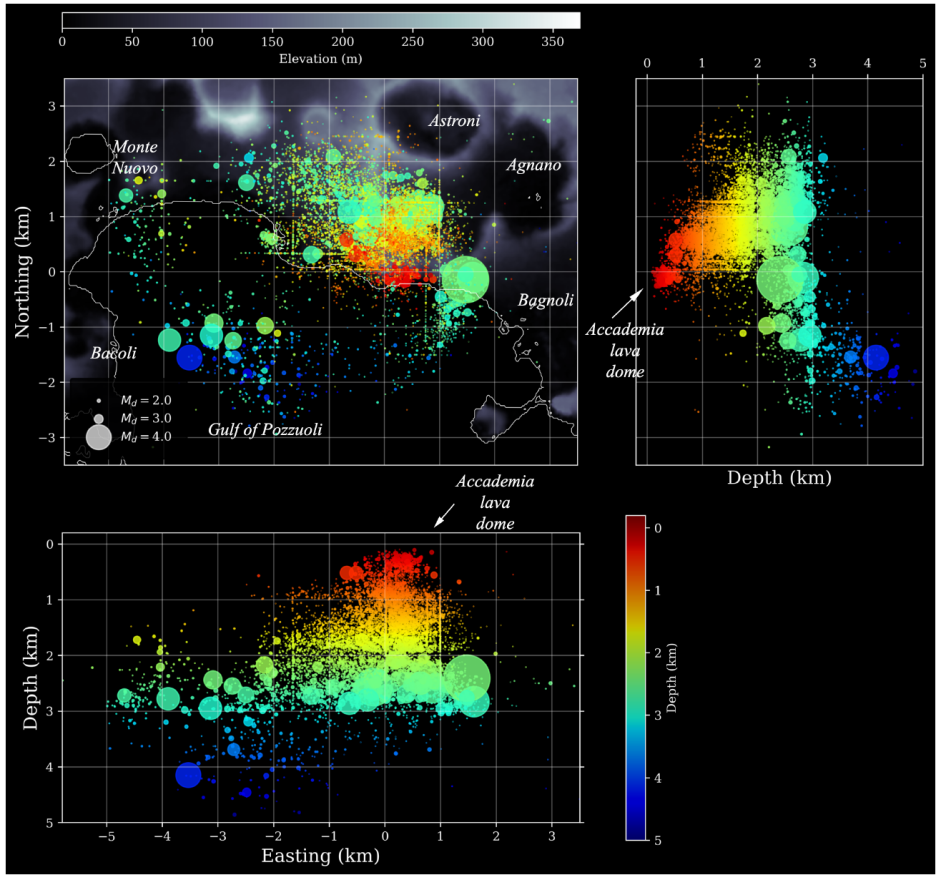
Partial S4A inference comparison on original CSOB observations, 20 May 2024 20:10:03–20:12:03 UTC. The caption's two-hour offset is interpreted literally; exact published alignment is unverified. Original weights: 5 P and 5 S; retrained weights: 6 P and 10 S. No labeled accuracy or retraining claim.

Missing: S4B — not reproduced

S4B: exact CMIS noise-window timing and reference labels are unavailable. Other calendar-window diagnostics are not inserted as that panel.

Figure S5: Routine-catalog locations

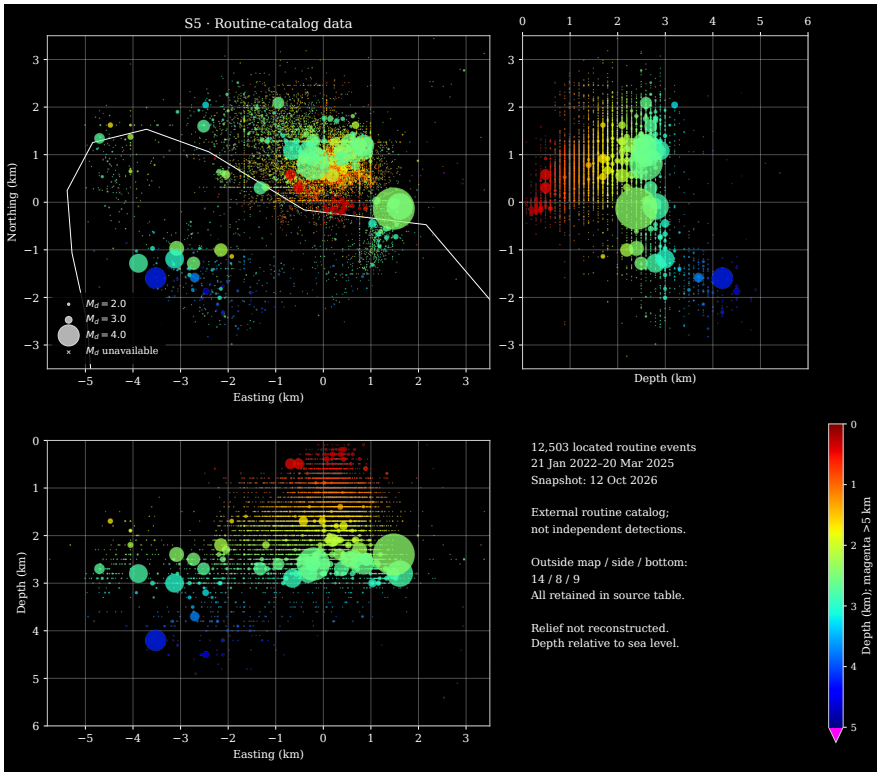
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 16.
DOI: 10.1126/science.adw9038.

Independent result

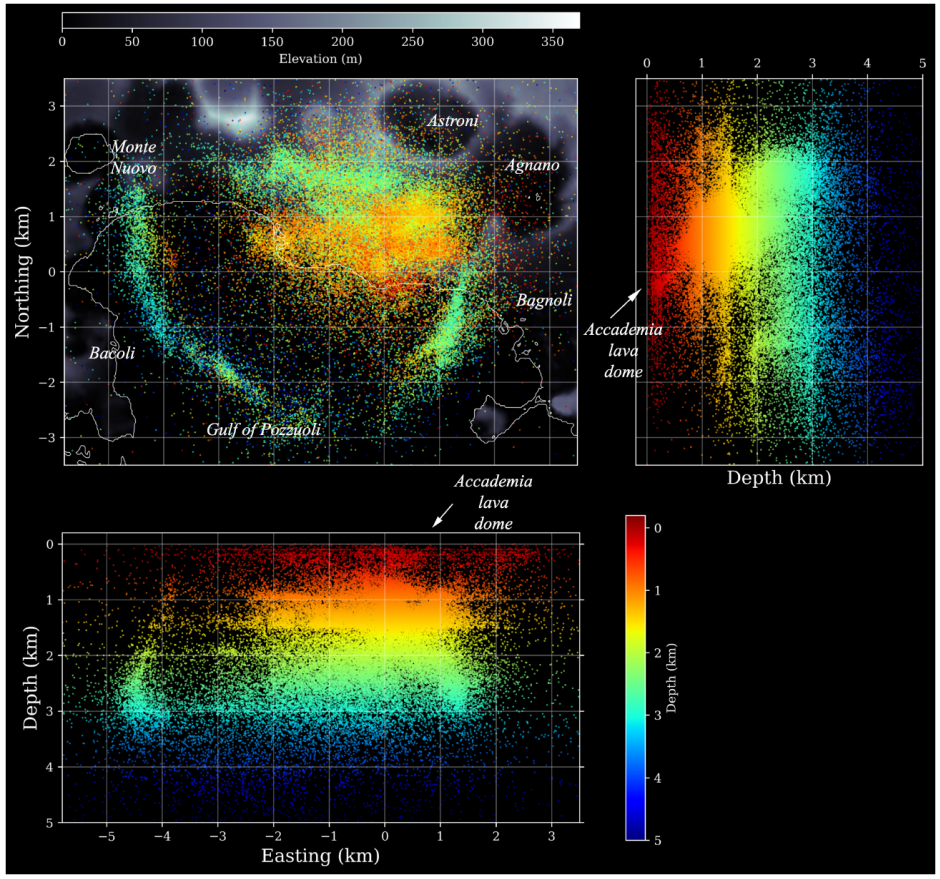
Panel S5, all views — partially reproduced



External routine-catalog data, independently selected and plotted: 12,503 located events, 21 January 2022–20 March 2025. This is 420 more than the published historical snapshot. No author-derived catalog is used to adjust that difference; these are not independent detections. Depth colors: fixed 0–5 km, red to blue; magenta >5 km, no data clipping. S5 limits/ticks are printed; 1A endpoints are unlabeled and 0–5 km is an explicit approximation. Circle area law is an explicit display assumption, not recovered exactly. Missing magnitudes are crosses. Routine events are external context.

Figure S6: Initial VELEST locations

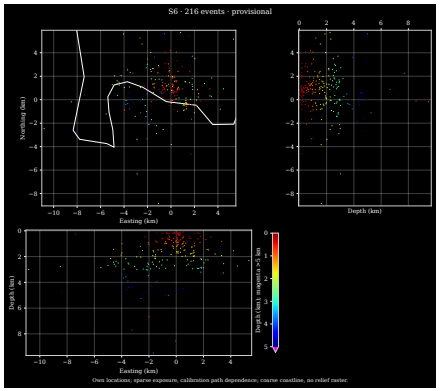
Published reference



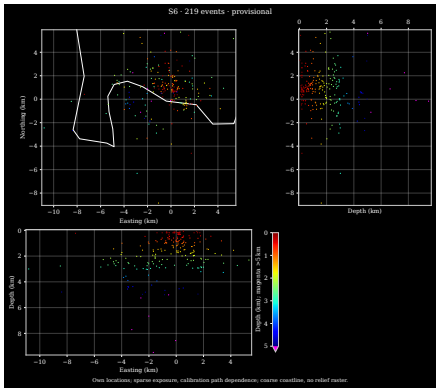
Tan et al. (2025), tanmodelsupp.pdf, p. 17.
DOI: 10.1126/science.adw9038.

Independent result

Panel S6; provisional numerical counterpart — partially reproduced



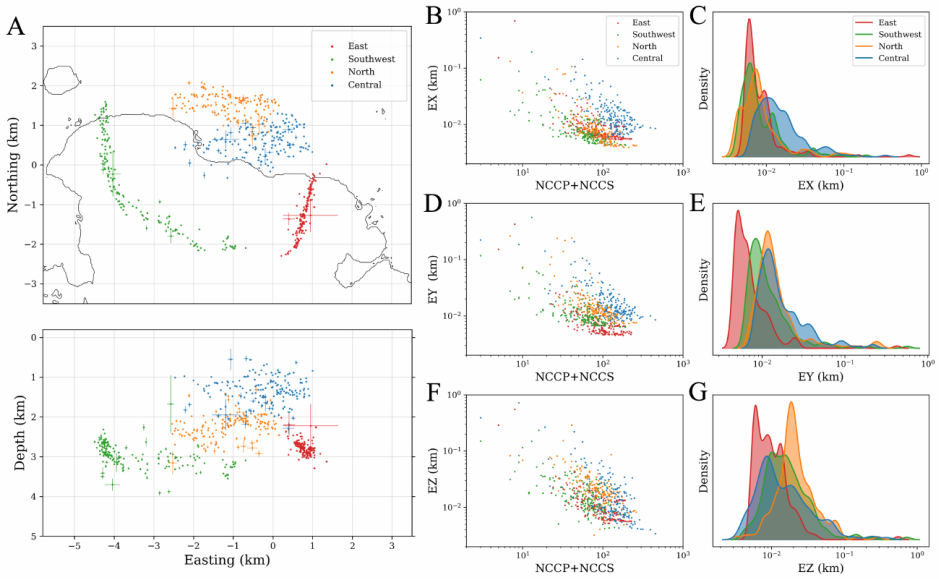
Panel Latest extended cohort: independently rerun, provisional — partially reproduced



216 and 219 own VELEST numerical-filter passes. Uniform symbols and fixed 0–5 km depth colors; magenta marks deeper events. Calibration remains unaccepted; no author locations or fitted model are plotted.

Figure S7: Location uncertainties

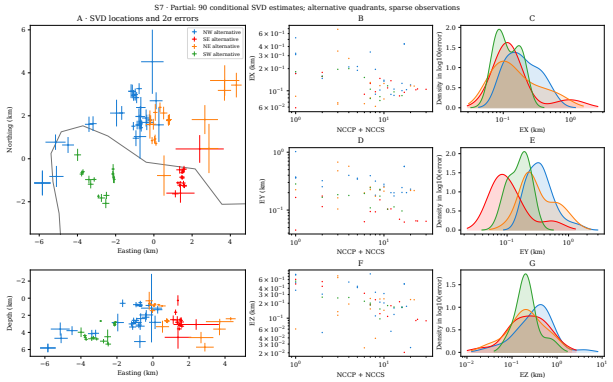
Published reference



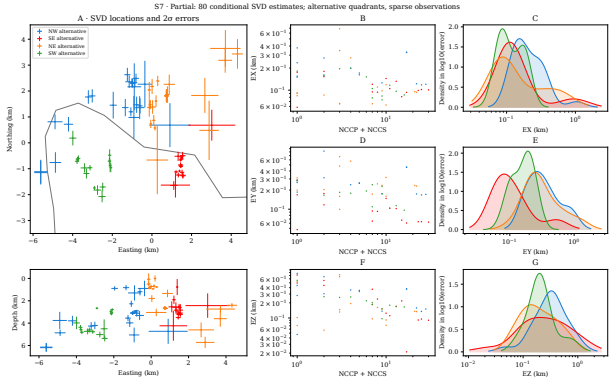
Tan et al. (2025), tanmodelsupp.pdf, p. 18.
DOI: 10.1126/science.adw9038.

Independent result

Panel S7; provisional numerical counterpart — partially reproduced



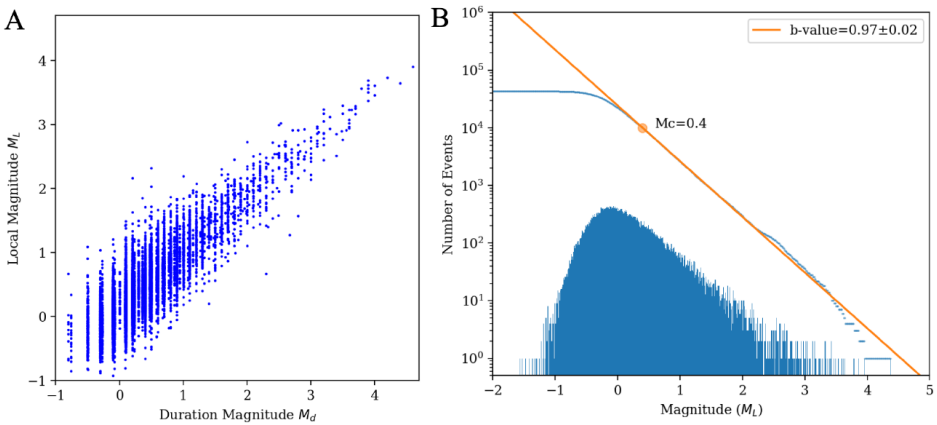
Panel Latest extended cohort: independently rerun, provisional — partially reproduced



90 and 80 conditional SVD solutions from alternative quadrants. Kilometer-scale errors are retained. Different subset counts do not establish improved absolute accuracy.

Figure S8: Magnitude calibration and frequency distribution

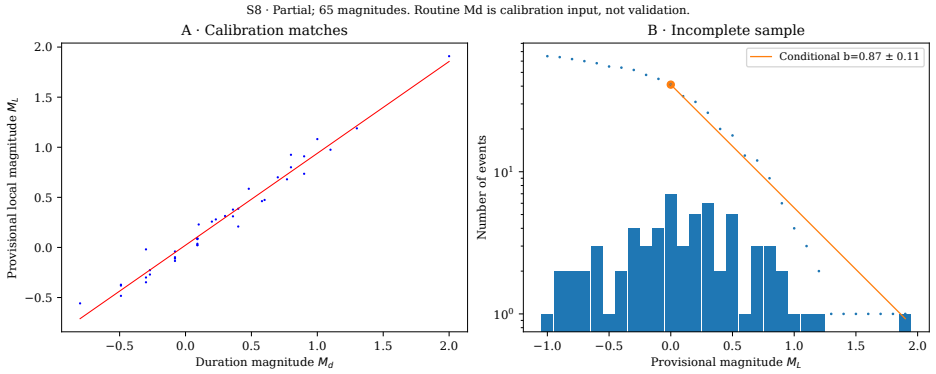
Published reference



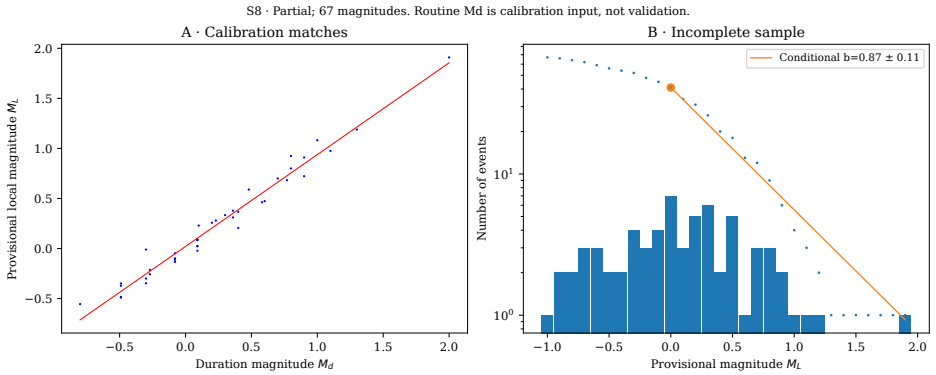
Tan et al. (2025), tanmodelsupp.pdf, p. 19.
DOI: 10.1126/science.adw9038.

Independent result

Panel S8A–B — partially reproduced



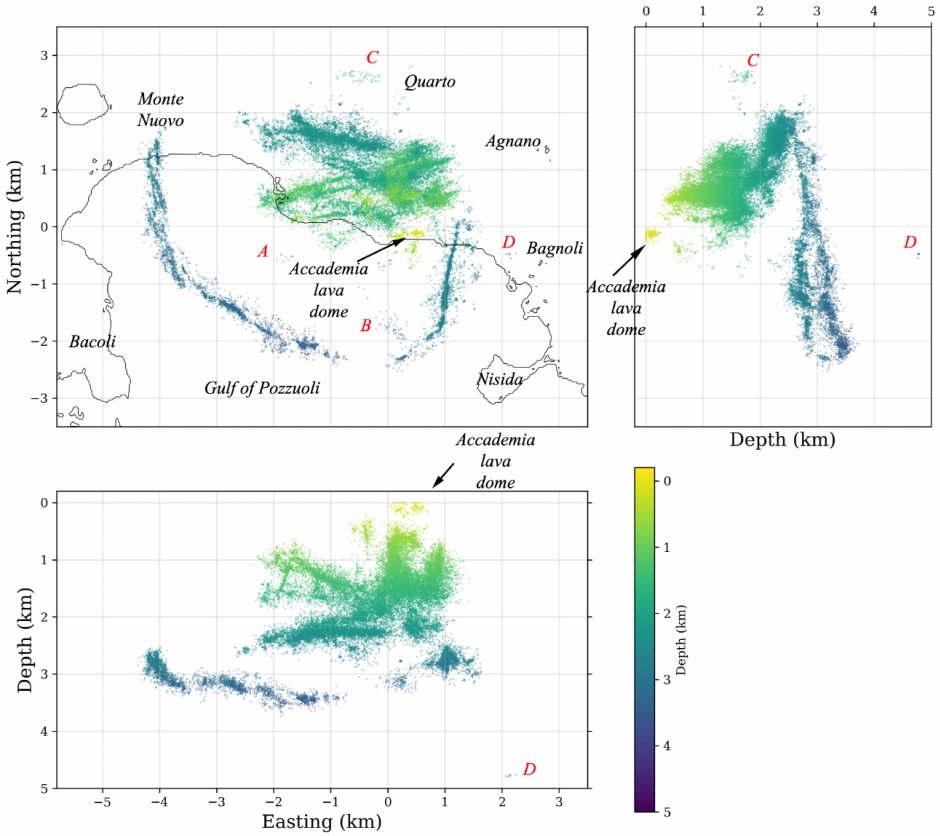
Panel Latest extended cohort: independently rerun, provisional — partially reproduced



65 and 67 independently inverted amplitude magnitudes. Routine duration magnitude is a calibration prior, not validation. Completeness and b describe selected, discontinuous samples.

Figure S9: Best-located earthquake subset

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 20.
DOI: 10.1126/science.adw9038.

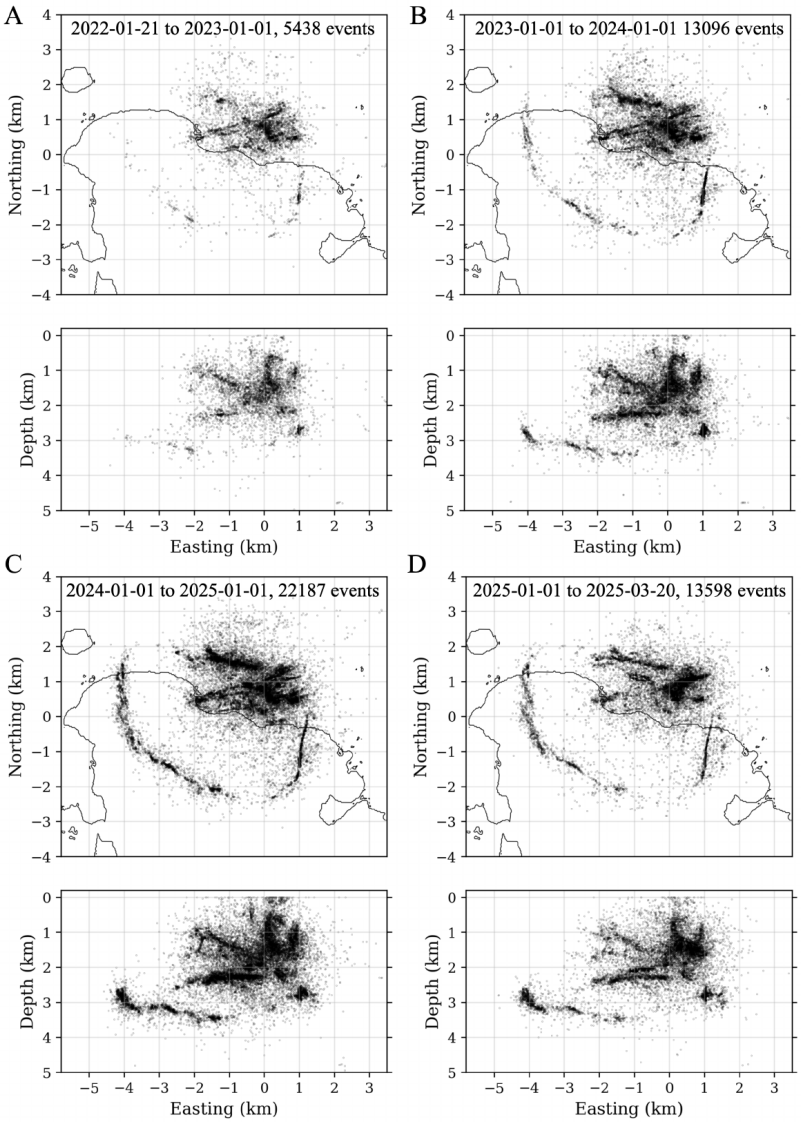
Independent result

Missing: S9/map, S9/northing-depth, S9/easting-depth — not reproduced

The documented NCCP+NCCS>100 selection was computed on the provisional relocated catalog and returned zero events. No qualified spatial result exists to plot. More waveform-linked neighboring events are needed; the threshold was not relaxed.

Figure S10: Seismicity by year

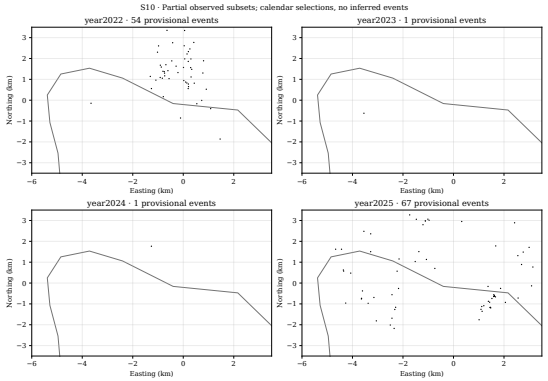
Published reference



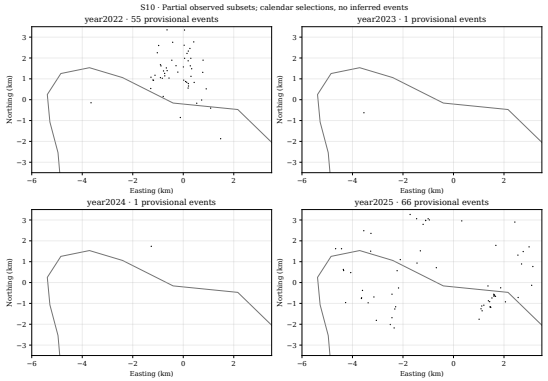
Tan et al. (2025), tanmodelsupp.pdf, p. 21.
DOI: 10.1126/science.adw9038.

Independent result

Panel S10; provisional numerical counterpart — partially reproduced



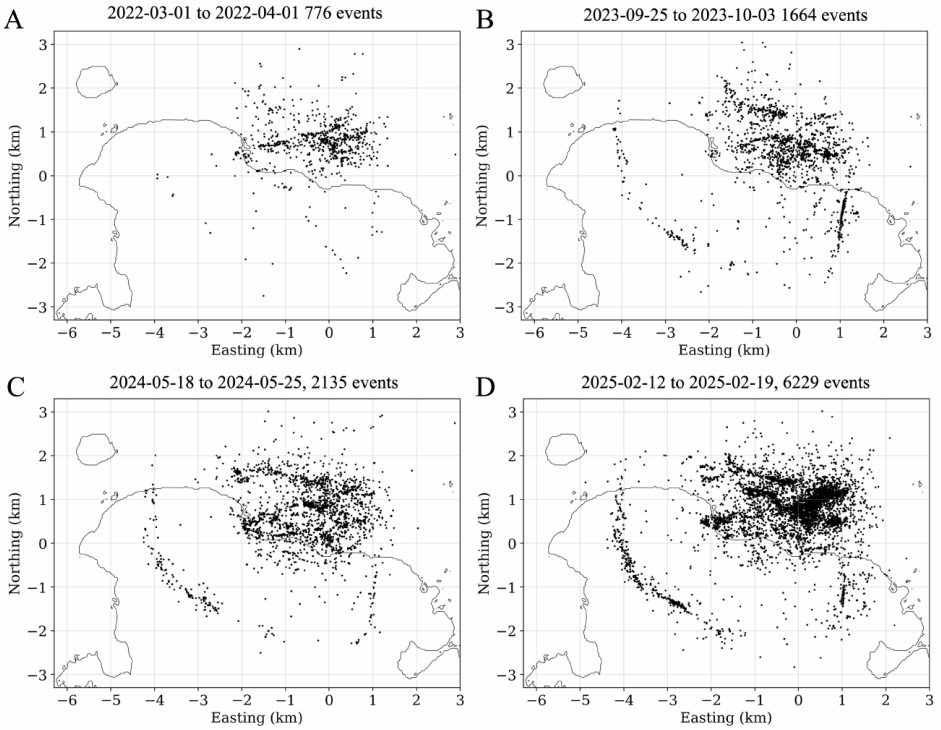
Panel Latest extended cohort: independently rerun, provisional — partially reproduced



Own trial locations by calendar year, with sparse coverage. Counts are not annual seismicity-rate estimates.

Figure S11: Four major swarms

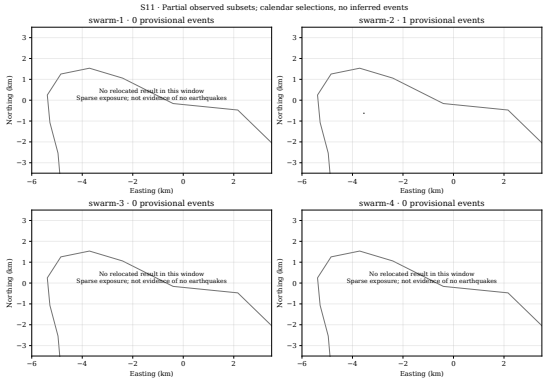
Published reference



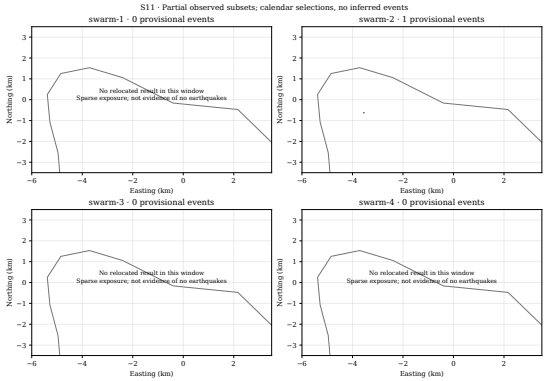
Tan et al. (2025), tanmodelsupp.pdf, p. 22.
DOI: 10.1126/science.adw9038.

Independent result

Panel S11; provisional numerical counterpart — partially reproduced



Panel Latest extended cohort: independently rerun, provisional — partially reproduced



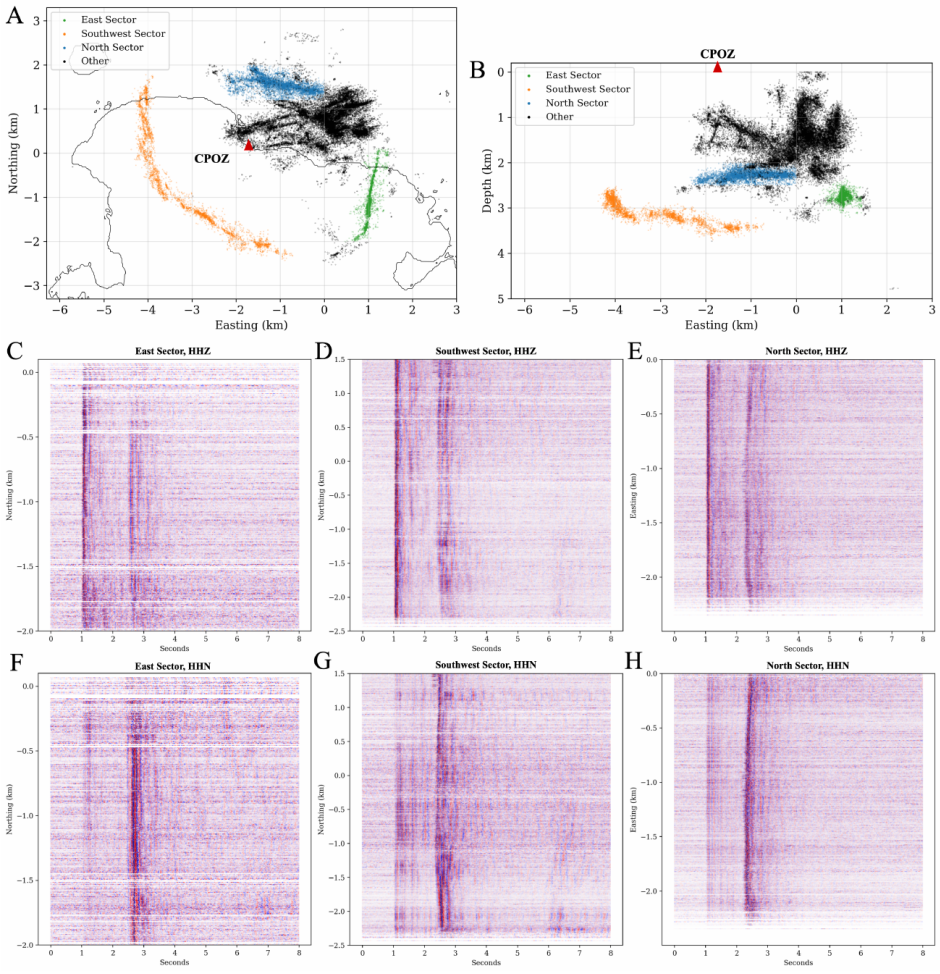
Only swarm 2 contains retained own events. The other written calendar windows have no qualifying numerical result in either observed subset.

Missing: S11/swarm-1, S11/swarm-3, S11/swarm-4 — not reproduced

S11/swarm-1, swarm-3 and swarm-4 have no surviving independently relocated events in their written calendar windows; sparse exposure, not evidence that no earthquakes occurred.

Figure S12: Ring sectors and CPOZ common-receiver gathers

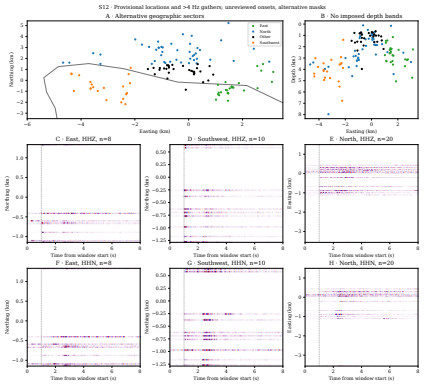
Published reference



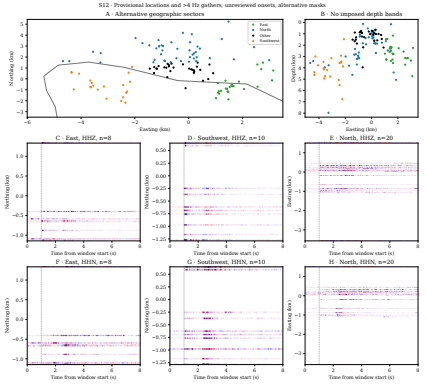
Tan et al. (2025), tanmodelsupp.pdf, p. 23.
DOI: 10.1126/science.adw9038.

Independent result

Panel S12; provisional numerical counterpart — partially reproduced



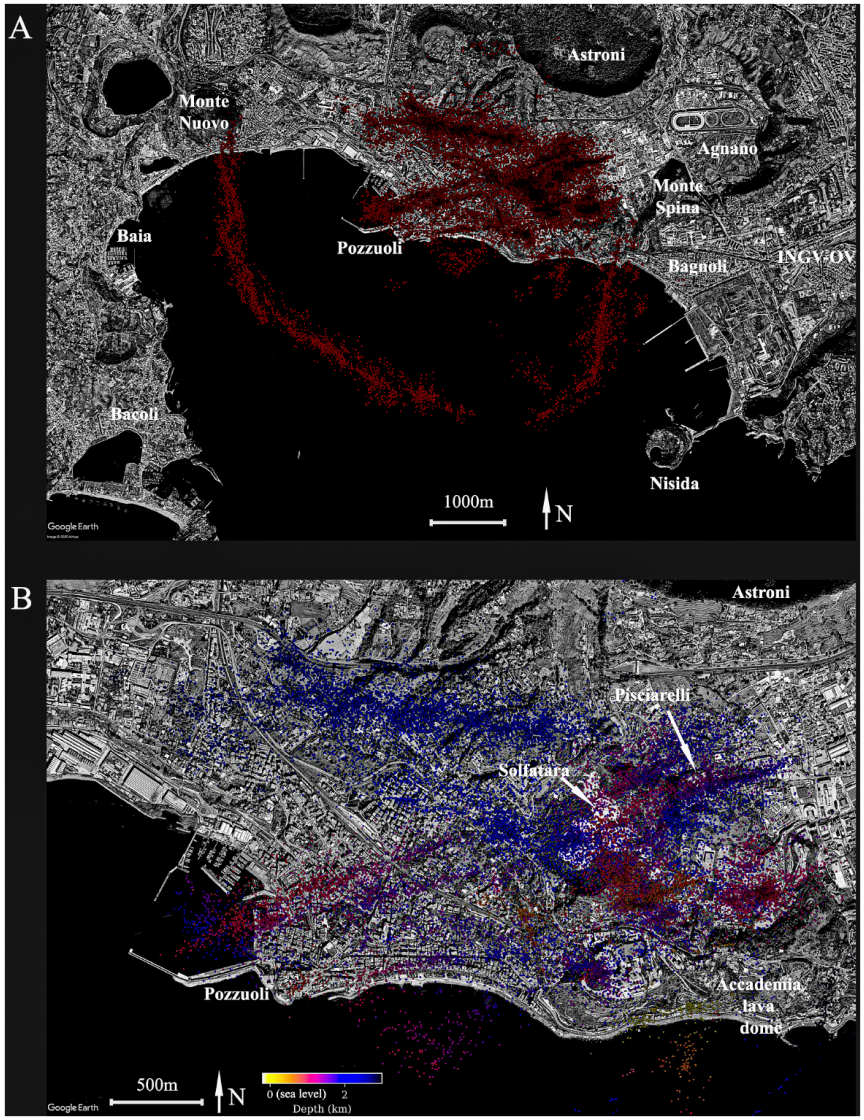
Panel Latest extended cohort: independently rerun, provisional — partially reproduced



118 and 122 original CPOZ component traces, aligned on own P picks and filtered above 4 Hz. Alternative geographic masks, individually normalized amplitudes and unreviewed onsets.

Figure S13: Relocated seismicity on satellite imagery

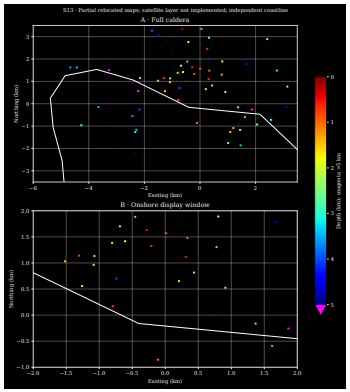
Published reference



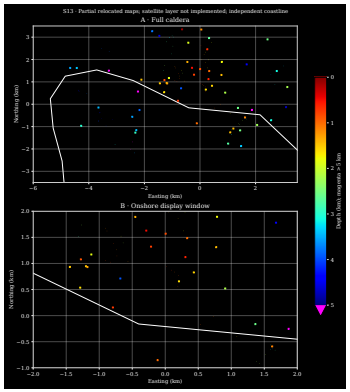
Tan et al. (2025), tanmodelsupp.pdf, p. 24.
DOI: 10.1126/science.adw9038.

Independent result

Panel S13; provisional numerical counterpart — partially reproduced



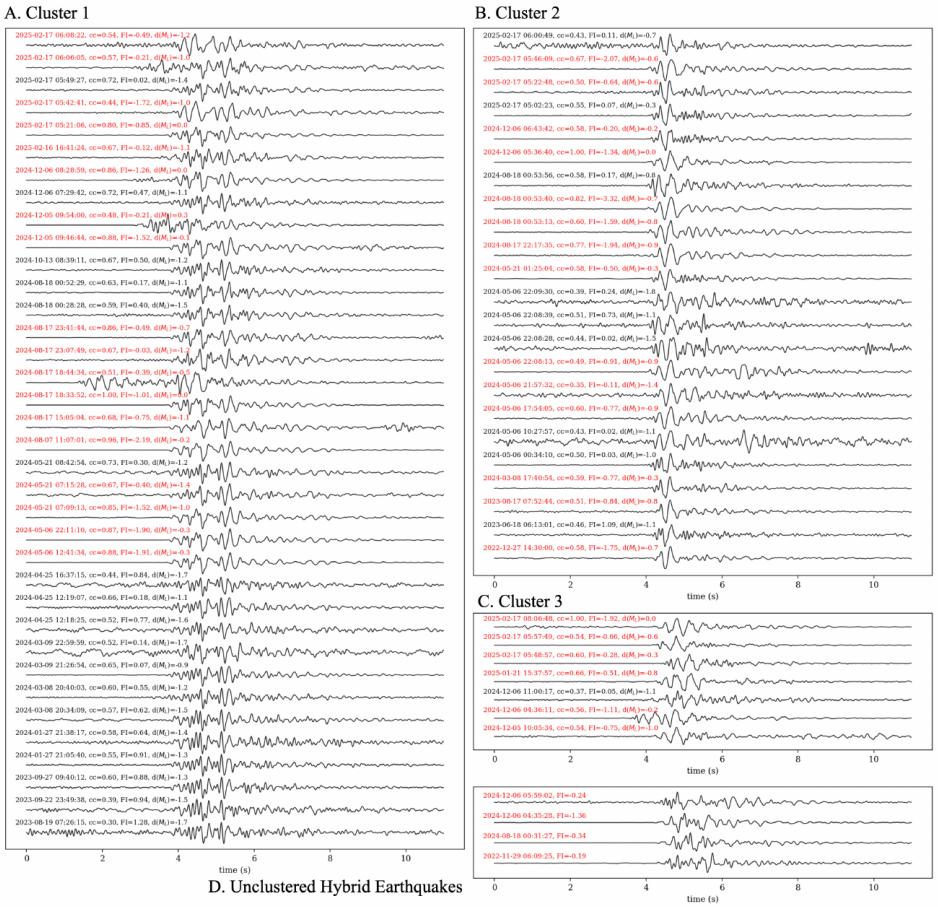
Panel Latest extended cohort: independently rerun, provisional — partially reproduced



Own provisional relocated coordinates on an independent coarse coastline. A satellite background is not implemented; no published pixels supply the result.

Figure S14: Hybrid-event clusters

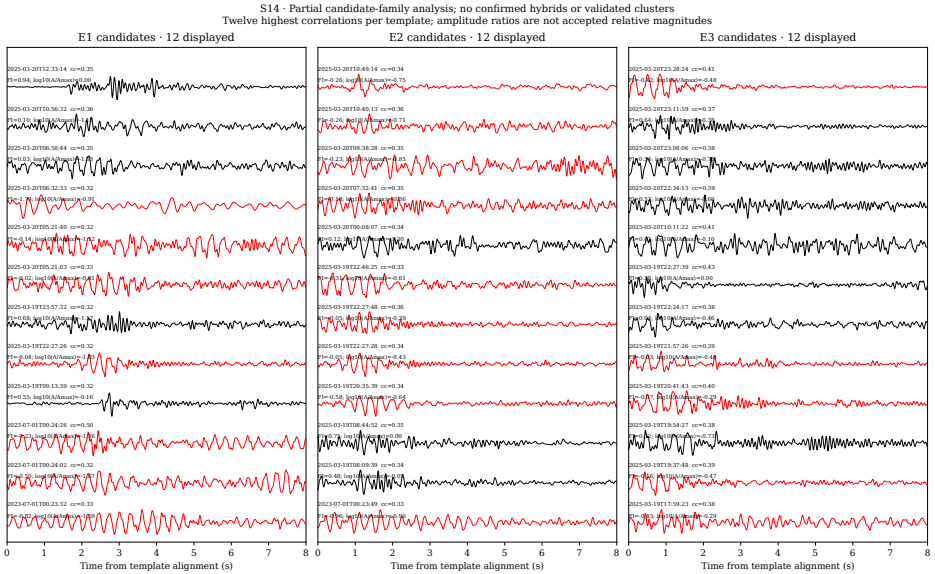
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 25.
DOI: 10.1126/science.adw9038.

Independent result

Panel Template-family candidates; not confirmed clusters — partially reproduced



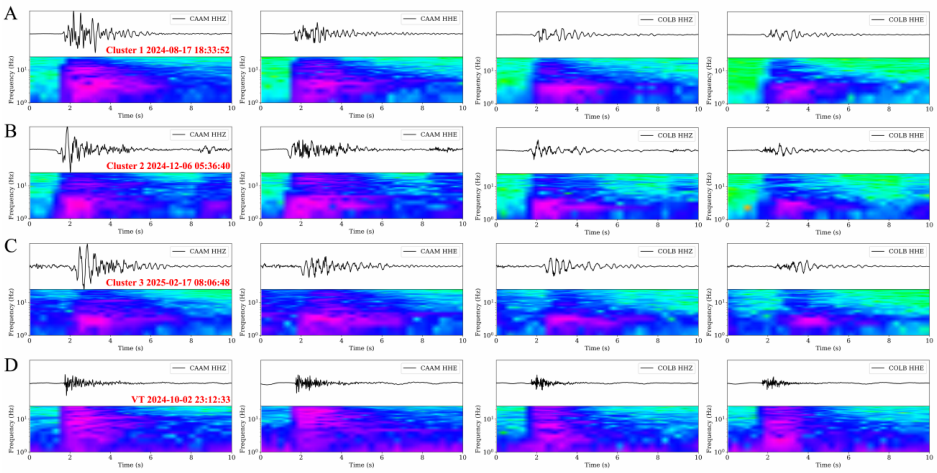
Measured waveforms, frequency indices and amplitude ratios for the twelve highest correlations per template. These candidates are not confirmed hybrids or validated clusters.

Missing: S14/unclustered — not reproduced

S14/unclustered: no independent unclustered hybrid sample or validated family assignment. The three-template threshold scan cannot itself supply that class.

Figure S15: Hybrid and shallow-event waveforms

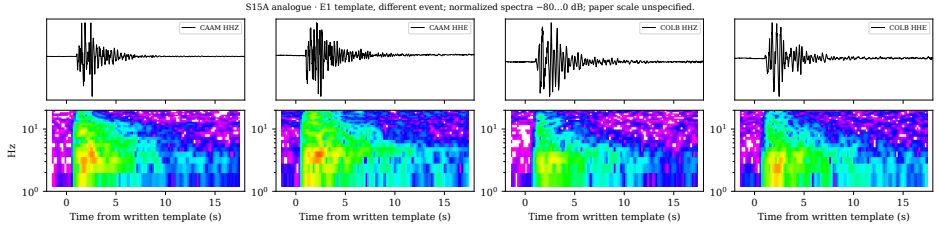
Published reference



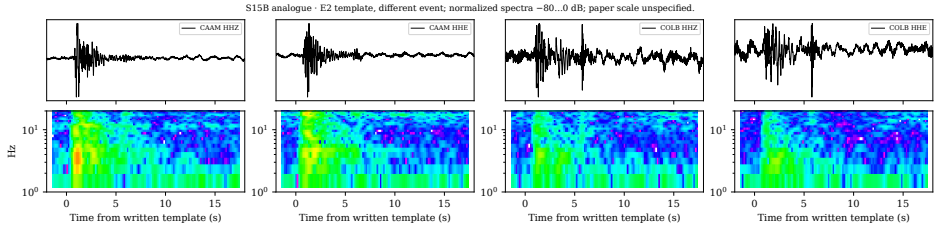
Tan et al. (2025), tanmodelsupp.pdf, p. 26.
DOI: 10.1126/science.adw9038.

Independent result

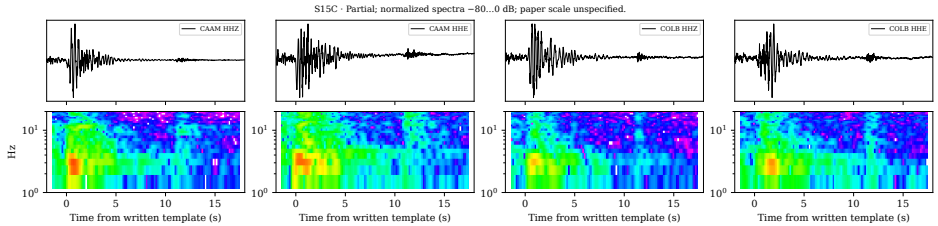
Panel S15A analogue: different documented E1 event — partially reproduced



Panel S15B analogue: different documented E2 event — partially reproduced



Panel S15C: documented E3 event — partially reproduced



A/B use explicitly different E1/E2 examples; C uses the documented E3 template. All traces and spectra were computed from original observations. No qualified D example is available.

Missing: S15D — not reproduced

S15D: no retained independently relocated event with depth <0.6 km south of 40.83 N in this cohort, hence no suitable west-Accademia extremely-shallow example. Incorrectly labeled examples are not substituted.